

APPENDIX B

(REQUIRED FOR OPEN-BOOK EXAMINATION)

DESIGN DATA

CHAIN DRIVE

TABLES:

B-1: Dimensions, Tensile Strengths, Allowable Loads

B-2: Service Factor

B-3: Multiple Strand Factor

B-4: Power Ratings

Source: Tsubakimoto Chain

Table B-1

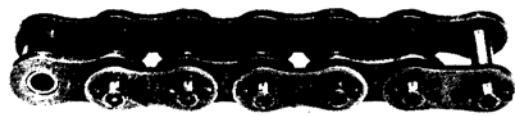
AMERICAN STANDARD ROLLER CHAINS-SINGLE STRAND

ANSI STANDARD ROLLER CHAINS

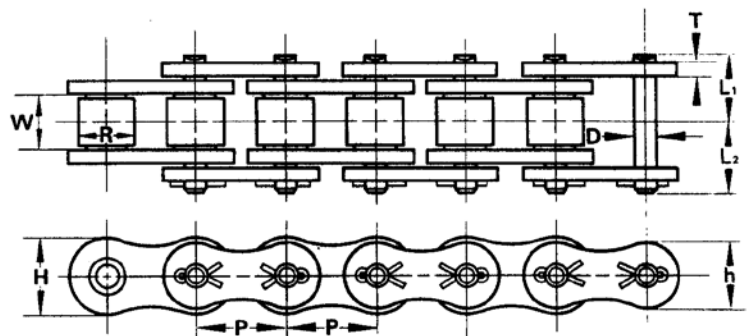
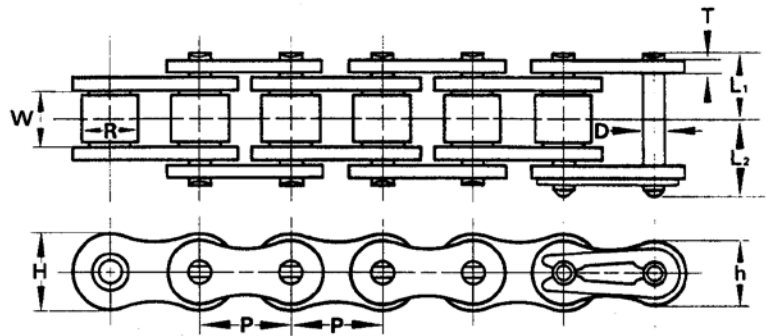
Single and Multiple Strand Standard Roller Chains conform to the ANSI (American National Standard Institute) Interchangeable with other chains conforming to ANSI Standards.



RIVETED TYPE



COTTERED TYPE



Upper dimension—inch
Lower dimension—mm

ANSI No.	Pitch	Roller Diam.	Width between roller link plates	Link Plate			Pin			Average Tensile Strength	Maximum Allowable Load	Approx. Weight	Number of Link per 10 Ft.	Length Per Carton Package (Ft.)
	P	R	W	Thickness	Height	Height	Diam.	From Pin head to C.L.	From Pin end to C.L.					
	P	R	W	T	H	h	D	L ₁	L ₂	Lbs./kg	Lbs./kg	Lbs./ft kg/mt		
25	1/4 6.35	.130 3.30	3/16 3.18	.030 0.75	.230 5.84	.199 5.05	.0905 2.31	.148 3.85	.179 4.80	1,050 480	140 65	.094 0.14	480	400
35	3/8 9.525	.200 5.08	3/16 4.78	.050 1.25	.354 9.0	.307 7.8	.141 3.59	.230 5.85	.270 6.85	2,500 1,150	480 220	.22 0.33	320	100
41	1/2 12.70	.306 7.77	1/4 6.38	.050 1.25	.386 9.8	.331 8.4	.141 3.59	.266 6.75	.313 7.95	2,600 1,200	500 230	.27 0.41	240	100
40	1/2 12.70	.312 7.94	3/16 7.95	.060 1.5	.472 12.0	.409 10.4	.156 3.97	.325 8.25	.392 9.95	4,250 1,950	810 370	.43 0.64	240	100
50	5/8 15.875	.400 10.16	3/8 9.53	.080 2.0	.591 15.0	.512 13.0	.200 5.09	.406 10.3	.472 12.0	7,050 3,200	1,400 650	.70 1.04	192	100
60	3/4 19.05	.469 11.91	1/2 12.70	.094 2.4	.713 18.1	.614 15.6	.234 5.96	.506 12.85	.600 15.25	9,900 4,500	1,950 900	1.03 1.53	160	100
80	1 25.40	.625 15.88	5/8 15.88	.125 3.2	.949 24.1	.819 20.8	.312 7.94	.640 16.25	.758 19.25	17,600 8,000	3,300 1,500	1.79 2.66	120	50
100	1 1/4 31.75	.750 19.05	3/4 19.05	.156 4.0	1.185 30.1	1.024 26.0	.375 9.54	.778 19.75	.900 22.85	26,400 12,000	5,060 2,300	2.68 3.99	96	50
120	1 1/2 38.10	.875 22.23	1 25.40	.187 4.8	1.425 36.2	1.228 31.2	.437 11.11	.980 24.9	1.138 28.9	37,400 17,000	6,800 3,100	3.98 5.93	80	20
140	1 3/4 44.45	1.000 25.40	1 25.40	.219 5.6	1.661 42.2	1.433 36.4	.500 12.71	1.059 26.9	1.248 31.7	48,500 22,000	9,000 4,100	5.03 7.49	68	20
160	2 50.80	1.125 28.58	1 1/4 31.75	.250 6.4	1.898 48.2	1.638 41.6	.562 14.29	1.254 31.85	1.451 36.85	60,600 27,500	11,900 5,400	6.79 10.10	60	20
180	2 1/4 57.15	1.406 35.71	1 1/2 35.72	.281 7.15	2.059 54.2	1.709 46.8	.687 17.46	1.404 35.65	1.671 42.45	80,400 36,500	13,600 6,200	9.04 13.45	54	10
200	2 1/2 63.50	1.562 39.69	1 1/2 38.10	.312 8.0	2.374 60.3	2.047 52.0	.781 19.85	1.535 39.0	1.764 44.8	103,600 47,000	16,000 7,300	11.04 16.49	48	10
240	3 76.20	1.875 47.63	1 3/4 47.63	.375 9.5	2.850 72.4	2.457 62.4	.937 23.81	1.886 47.9	2.185 55.5	152,100 69,000	22,200 10,100	16.46 24.5	40	10

Table B-2**Service Factors for Roller Chain Drives**

Type of Driven Load	Type of Input Power		
	Internal Combustion Engine with Hydraulic Drive	Electric Motor or Turbine	Internal Combustion Engine with Mechanical Drive
Smooth	1.0	1.0	1.2
Moderate Shock	1.2	1.3	1.4
Heavy Shock	1.4	1.5	1.7

Load Classifications		
Smooth Load	Moderate Shock Load	Heavy Shock Load
Agitators—Pure liquid Conveyors—Uniformly loaded or fed (apron, assembly, belt, flight, oven, screw) Fans—Centrifugal and light, small diameter Line shafts—Light Machines—All types with uniform non-reversing loads Screens—Rotary (uniformly fed), traveling water intake Sewage disposal equipment—Inside service (uniformly fed)	Clay working machinery—Pug mills Conveyors—Heavy duty and NOT uniformly loaded (apron, assembly, belt, bucket, flight, oven, screw) Cranes and hoists—Medium duty, skip hoists (travel motion and trolley motion) Dredges—Cable, reel and conveyor Food industry—Beet slicers, dough mixers, meat grinders Grinders Laundry industry—Washers, tumblers Line shafts—Heavy service Machine tools—Main and auxiliary drives Machine—All types with moderate shock and nonreversing loads Screens—Rotary (stone or gravel) Textile industry—Calenders, dyeing machinery, mangles, nappers, soapers, spinners, tenter frames	Clay working machinery—Brick press, briquetting machinery Conveyors—Reciprocating and shaker Cranes and Hoists—Heavy duty, including logging, lumbering, and rotary drilling rigs Dredges—Cutter head drives, jig drives, screen drives Hammer mills Machine tools—Punch press, shears, plate planers Machines—All types with severe impact shock loads or speed variations, and reversing service Metal mills—Draw bench, forming machines, slitters, small rolling mill drives, wire drawing or flattening Mills—(rotary type) ball, cement kilns, rod mills, tumbling mills Paper industry—Mixers, calendars, rubber mill sheeters Textile industry—Carding machinery

Table B-3 Multiple Strand Factors

Number of Strands	Multiple Strand Factor
1	1.0
2	1.7
3	2.5
4	3.3
5	3.9
6	4.6

Table B-4

MAXIMUM KILOWATT RATINGS

Number of Roller Chain Strands	Multi-Strand Factor
2	1.7
3	2.5
4	3.3
5	3.9
6	4.6

Lubricating Methods	
A	Manual lubrication or drip lubrication.
B	Oil bath lubrication or lubrication by sling disc
C	Lubrication using a pump

NO. 25 MAXIMUM KILOWATT RATINGS

No. of Teeth Small Spkt.	Maximum r/min — Small Sprocket																									
	50	100	300	500	700	900	1200	1500	1800	2100	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	10000	
	Lubrication System																									
	A													B												
9	0.02	0.03	0.08	0.13	0.18	0.23	0.30	0.36	0.43	0.49	0.57	0.67	0.78	0.76	0.64	0.54	0.47	0.42	0.37	0.33	0.30	0.27	0.25	0.22	0.19	
10	0.02	0.04	0.10	0.15	0.20	0.26	0.33	0.41	0.48	0.55	0.64	0.76	0.87	0.89	0.75	0.64	0.55	0.48	0.43	0.39	0.35	0.31	0.29	0.26	0.22	
11	0.02	0.04	0.11	0.17	0.23	0.28	0.37	0.45	0.53	0.61	0.71	0.84	0.96	1.03	0.87	0.74	0.64	0.56	0.50	0.45	0.40	0.37	0.34	0.31	0.26	
12	0.02	0.04	0.12	0.18	0.25	0.31	0.40	0.49	0.58	0.67	0.78	0.92	1.06	1.17	0.98	0.84	0.72	0.64	0.57	0.51	0.46	0.42	0.38	0.35	0.30	
13	0.03	0.05	0.13	0.20	0.27	0.34	0.44	0.54	0.63	0.73	0.85	1.00	1.15	1.30	1.11	0.95	0.82	0.72	0.64	0.57	0.51	0.47	0.43	0.40	0.34	
14	0.03	0.05	0.14	0.22	0.29	0.37	0.48	0.58	0.69	0.79	0.92	1.09	1.25	1.41	1.24	1.06	0.92	0.81	0.72	0.64	0.57	0.52	0.48	0.44	0.37	
15	0.03	0.05	0.15	0.23	0.32	0.40	0.51	0.63	0.74	0.85	0.99	1.17	1.35	1.52	1.37	1.17	1.01	0.90	0.79	0.71	0.64	0.58	0.53	0.48	0.42	
16	0.03	0.06	0.16	0.25	0.34	0.43	0.55	0.67	0.79	0.91	1.07	1.26	1.44	1.63	1.51	1.29	1.12	0.98	0.87	0.78	0.70	0.64	0.58	0.54	0.46	
17	0.03	0.06	0.17	0.27	0.36	0.45	0.59	0.72	0.85	0.97	1.14	1.34	1.54	1.74	1.66	1.42	1.22	1.07	0.96	0.85	0.77	0.70	0.64	0.59	0.50	
18	0.04	0.07	0.18	0.28	0.39	0.48	0.63	0.76	0.90	1.03	1.21	1.43	1.64	1.85	1.81	1.54	1.34	1.17	1.04	0.93	0.84	0.76	0.69	0.64	0.54	
19	0.04	0.07	0.19	0.30	0.41	0.51	0.66	0.81	0.96	1.10	1.28	1.51	1.74	1.96	1.95	1.67	1.45	1.27	1.13	1.01	0.91	0.83	0.75	0.69	0.59	
20	0.04	0.07	0.20	0.32	0.43	0.54	0.70	0.88	1.01	1.16	1.36	1.60	1.84	2.07	2.11	1.81	1.57	1.37	1.22	1.09	0.98	0.90	0.81	0.75	0.64	
21	0.04	0.08	0.21	0.34	0.45	0.57	0.74	0.90	1.06	1.22	1.43	1.69	1.94	2.18	2.27	1.94	1.69	1.48	1.31	1.17	1.06	0.96	0.87	0.81	0.69	
22	0.04	0.08	0.22	0.35	0.48	0.60	0.78	0.95	1.12	1.29	1.50	1.77	2.04	2.30	2.44	2.08	1.81	1.58	1.40	1.26	1.13	1.03	0.94	0.87	0.74	
23	0.05	0.09	0.23	0.37	0.50	0.63	0.82	1.00	1.17	1.35	1.58	1.86	2.14	2.41	2.61	2.22	1.93	1.69	1.50	1.34	1.21	1.10	1.01	0.93	0.79	
24	0.05	0.09	0.25	0.39	0.53	0.66	0.85	1.04	1.23	1.41	1.65	1.95	2.24	2.52	2.78	2.37	2.06	1.81	1.60	1.43	1.29	1.17	1.07	0.98	0.84	
25	0.05	0.10	0.26	0.41	0.55	0.69	0.89	1.09	1.28	1.48	1.73	2.03	2.34	2.64	2.93	2.52	2.19	1.92	1.70	1.52	1.37	1.25	1.14	1.04	0.90	
26	0.05	0.10	0.27	0.42	0.57	0.72	0.93	1.14	1.34	1.54	1.80	2.12	2.44	2.75	3.06	2.68	2.32	2.04	1.81	1.61	1.45	1.32	1.21	1.11	0.95	
28	0.06	0.11	0.29	0.46	0.62	0.78	1.01	1.23	1.45	1.67	1.95	2.30	2.64	2.98	3.31	2.99	2.59	2.28	2.01	1.81	1.63	1.48	1.35	1.24	1.06	
30	0.06	0.12	0.31	0.49	0.67	0.84	1.09	1.33	1.56	1.80	2.10	2.48	2.85	3.21	3.57	3.32	2.87	2.52	2.24	2.00	1.81	1.64	1.50	1.37	1.17	
32	0.07	0.12	0.33	0.53	0.72	0.90	1.16	1.42	1.68	1.93	2.25	2.66	3.05	3.44	3.83	3.66	3.17	2.78	2.46	2.21	1.99	1.81	1.65	1.51	1.29	
35	0.07	0.14	0.37	0.58	0.79	0.99	1.28	1.57	1.85	2.12	2.48	2.93	3.36	3.79	4.21	4.18	3.62	3.18	2.82	2.52	2.28	2.07	1.89	1.73	1.48	
40	0.08	0.16	0.43	0.67	0.91	1.14	1.48	1.81	2.13	2.45	2.87	3.38	3.88	4.38	4.87	5.10	4.42	3.89	3.45	3.08	2.78	2.52	2.31	2.11	1.81	
45	0.10	0.18	0.48	0.77	1.04	1.30	1.68	2.06	2.42	2.78	3.26	3.84	4.41	4.97	5.53	6.08	5.28	4.63	4.11	3.68	3.32	3.01	2.75	2.52	2.16	

Notes: 1. Multiply the value given above by the multi-strand factor in order to obtain the transmission kW of multi-strand chain.

Table B-4

Multi-Strand Factor	
Number of Roller Chain Strands	Multi-Strand Factor
2	1.7
3	2.5
4	3.3
5	3.9
6	4.6

Lubricating Methods	
A	Manual lubrication or drip lubrication
B	Oil bath lubrication or lubrication by sling disc
C	Lubrication using a pump

NO. 35 MAXIMUM KILOWATT RATINGS

No. of Teeth Small Splt.	Maximum r/min — Small Sprocket																									
	50	100	300	500	700	900	1200	1500	1800	2100	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	10000	
	Lubrication System																									
	A								B																C	
9	0.07	0.14	0.37	0.59	0.80	1.00	1.30	1.59	1.88	2.15	2.11	1.82	1.29	1.05	0.88	0.75	0.66	0.57	0.51	0.46	0.41	0.37	0.34	0.31	0.27	
10	0.08	0.16	0.42	0.66	0.90	1.13	1.46	1.78	2.10	2.41	2.50	1.90	1.51	1.23	1.04	0.88	0.77	0.67	0.60	0.53	0.48	0.43	0.40	0.37	0.31	
11	0.09	0.17	0.46	0.74	1.00	1.25	1.62	1.98	2.33	2.68	2.88	2.18	1.73	1.42	1.18	1.01	0.88	0.77	0.68	0.61	0.55	0.50	0.45	0.42	0.36	
12	0.10	0.19	0.51	0.81	1.09	1.37	1.78	2.17	2.56	2.94	3.28	2.50	1.98	1.62	1.36	1.16	1.01	0.88	0.78	0.70	0.63	0.57	0.52	0.48	0.41	
13	0.11	0.21	0.58	0.88	1.19	1.49	1.94	2.37	2.79	3.20	3.70	2.80	2.23	1.83	1.53	1.30	1.12	0.99	0.87	0.78	0.70	0.64	0.58	0.54	0.46	
14	0.12	0.22	0.60	0.95	1.29	1.62	2.10	2.56	3.02	3.47	4.06	3.14	2.49	2.03	1.71	1.48	1.27	1.11	0.98	0.88	0.79	0.72	0.65	0.60	0.51	
15	0.13	0.24	0.65	1.03	1.39	1.74	2.26	2.76	3.26	3.74	4.38	3.47	2.78	2.26	1.89	1.62	1.40	1.23	1.09	0.98	0.88	0.80	0.73	0.67	0.57	
16	0.14	0.26	0.70	1.10	1.49	1.87	2.42	2.96	3.49	4.01	4.69	3.80	3.04	2.49	2.09	1.78	1.54	1.36	1.20	1.07	0.97	0.88	0.80	0.74	0.62	
17	0.15	0.28	0.74	1.18	1.59	2.00	2.59	3.16	3.73	4.28	5.01	4.17	3.33	2.73	2.28	1.95	1.69	1.48	1.32	1.18	1.06	0.96	0.88	0.80	0.69	
18	0.16	0.29	0.79	1.25	1.69	2.12	2.75	3.36	3.96	4.55	5.33	4.54	3.63	2.97	2.49	2.12	1.84	1.62	1.43	1.28	1.15	1.05	0.96	0.88	0.75	
19	0.17	0.31	0.84	1.33	1.80	2.25	2.92	3.57	4.20	4.83	5.65	4.92	3.94	3.22	2.70	2.30	2.00	1.75	1.56	1.39	1.25	1.14	1.03	0.95	0.81	
20	0.18	0.33	0.89	1.40	1.90	2.38	3.08	3.77	4.44	5.10	5.97	5.32	4.25	3.48	2.91	2.49	2.16	1.89	1.68	1.50	1.36	1.23	1.12	1.03	0.88	
21	0.19	0.35	0.93	1.48	2.00	2.51	3.25	3.97	4.68	5.38	6.29	5.72	4.57	3.74	3.14	2.68	2.32	2.03	1.80	1.62	1.46	1.32	1.21	1.11	0.95	
22	0.20	0.37	0.98	1.55	2.10	2.64	3.42	4.18	4.92	5.66	6.62	6.20	4.91	4.01	3.36	2.87	2.49	2.18	1.94	1.74	1.56	1.42	1.30	1.19	1.01	
23	0.21	0.38	1.03	1.63	2.21	2.77	3.59	4.38	5.17	5.93	6.94	6.62	5.26	4.30	3.60	3.08	2.67	2.34	2.08	1.86	1.68	1.52	1.39	1.28	1.09	
24	0.21	0.40	1.08	1.71	2.31	2.90	3.75	4.59	5.41	6.21	7.27	7.06	5.59	4.57	3.84	3.27	2.83	2.49	2.21	1.97	1.78	1.62	1.48	1.36	1.15	
25	0.22	0.42	1.13	1.78	2.42	3.03	3.92	4.80	5.66	6.49	7.60	7.51	5.96	4.88	4.09	3.49	3.02	2.66	2.36	2.10	1.90	1.72	1.57	1.45	1.23	
26	0.23	0.44	1.18	1.86	2.52	3.16	4.09	5.00	5.90	6.77	7.93	7.96	6.31	5.16	4.33	3.70	3.21	2.81	2.49	2.23	2.01	1.83	1.67	1.53	1.30	
28	0.25	0.47	1.27	2.02	2.73	3.42	4.43	5.42	6.39	7.34	8.59	8.90	7.07	5.78	4.84	4.14	3.59	3.15	2.79	2.50	2.25	2.04	1.87	1.72	1.48	
30	0.27	0.51	1.37	2.17	2.94	3.69	4.78	5.84	6.88	7.91	9.25	9.87	7.83	6.38	5.35	4.58	3.97	3.48	3.09	2.76	2.49	2.26	2.06	1.89	1.62	
32	0.29	0.55	1.47	2.33	3.15	3.95	5.12	6.26	7.38	8.48	9.92	10.9	8.58	7.04	5.90	5.04	4.37	3.83	3.40	3.04	2.74	2.49	2.27	2.09	0	
35	0.32	0.60	1.62	2.57	3.47	4.36	5.64	6.90	8.13	9.34	10.9	12.4	9.58	8.05	6.78	5.75	5.00	4.38	3.89	3.48	3.14	2.85	2.60	2.39	0	
40	0.37	0.70	1.87	2.96	4.01	5.03	6.52	7.97	9.39	10.88	12.6	14.9	12.0	9.85	8.28	7.05	6.11	5.36	4.75	4.25	3.83	3.48	0			
45	0.42	0.79	2.13	3.37	4.56	5.71	7.40	9.05	10.7	12.3	14.3	16.9	14.4	11.8	9.85	8.43	7.30	6.41	5.69	5.09	0					

NO. 40 MAXIMUM KILOWATT RATINGS

No. of Teeth Small Spkt.	Maximum r/min — Small Sprocket																									
	10	25	50	100	200	300	400	500	700	900	1000	1200	1400	1600	1800	2100	2400	2700	3000	3500	4000	5000	6000	7000	8000	
	Lubrication System																									
	A												B												C	
9	0.04	0.09	0.17	0.31	0.58	0.83	1.08	1.32	1.78	2.23	2.48	2.89	3.32	3.07	2.57	2.04	1.67	1.40	1.19	0.95	0.78	0.58	0.43	0.34		
10	0.04	0.10	0.19	0.35	0.65	0.93	1.21	1.47	2.00	2.50	2.75	3.24	3.72	3.62	3.01	2.39	1.96	1.64	1.40	1.11	0.91	0.65	0.49	0.40		
11	0.05	0.11	0.21	0.38	0.72	1.03	1.34	1.63	2.21	2.77	3.05	3.59	4.13	4.16	3.48	2.76	2.26	1.90	1.60	1.28	1.05	0.75	0.57	0.46		
12	0.05	0.12	0.23	0.42	0.79	1.13	1.47	1.79	2.43	3.05	3.35	3.95	4.53	4.74	3.96	3.15	2.57	2.16	1.84	1.46	1.19	0.85	0.65	0.51		
13	0.06	0.13	0.25	0.46	0.86	1.24	1.60	1.96	2.65	3.32	3.65	4.30	4.94	5.34	4.47	3.55	2.90	2.43	2.08	1.65	1.35	0.96	0.73	0.58		
14	0.06	0.14	0.27	0.50	0.93	1.34	1.73	2.12	2.87	3.60	3.96	4.66	5.36	5.97	5.00	3.96	3.25	2.72	2.32	1.84	1.51	1.08	0.82	0.65		
15	0.07	0.15	0.29	0.54	1.00	1.44	1.87	2.28	3.09	3.88	4.26	5.02	5.77	6.51	5.54	4.39	3.60	3.01	2.57	2.04	1.67	1.19	0.91	0.72		
16	0.07	0.17	0.31	0.58	1.07	1.55	2.00	2.45	3.32	4.16	4.57	5.39	6.19	6.98	6.10	4.84	3.96	3.32	2.84	2.25	1.84	1.32	1.00	0.80		
17	0.08	0.18	0.33	0.61	1.15	1.65	2.14	2.61	3.54	4.44	4.88	5.75	6.61	7.45	6.69	5.30	4.34	3.64	3.11	2.47	2.02	1.45	1.10	0.87		
18	0.08	0.19	0.35	0.65	1.22	1.76	2.28	2.78	3.76	4.72	5.19	6.12	7.03	7.92	7.28	5.78	4.73	3.96	3.39	2.69	2.21	1.57	1.19	0.95		
19	0.09	0.20	0.37	0.69	1.29	1.86	2.41	2.95	3.99	5.00	5.50	6.48	7.45	8.40	7.83	6.27	5.13	4.30	3.67	2.92	2.33	1.71	1.30	1.03		
20	0.09	0.21	0.39	0.73	1.37	1.97	2.55	3.12	4.22	5.29	5.82	6.85	7.87	8.88	8.28	6.77	5.54	4.64	3.96	3.15	2.57	1.84	1.40	1.11		
21	0.10	0.22	0.41	0.77	1.44	2.07	2.69	3.29	4.45	5.58	6.13	7.22	8.30	9.36	9.24	7.28	5.96	5.00	4.27	3.39	2.77	1.98	1.51	1.19		
22	0.10	0.23	0.43	0.81	1.51	2.18	2.83	3.45	4.68	5.86	6.45	7.60	8.73	9.84	9.66	7.83	6.39	5.36	4.57	3.63	2.97	2.13	1.62	1.28		
23	0.11	0.24	0.46	0.85	1.59	2.29	2.96	3.62	4.91	6.15	6.76	7.97	9.15	10.3	10.5	8.38	6.83	5.73	4.89	3.88	3.18	2.28	1.73	1.37		
24	0.11	0.26	0.48	0.89	1.66	2.40	3.10	3.79	5.14	6.44	7.08	8.34	9.59	10.8	11.2	8.88	7.28	6.10	5.22	4.13	3.39	2.42	1.84	1.46		
25	0.12	0.27	0.50	0.93	1.74	2.50	3.24	3.97	5.37	6.73	7.40	8.72	10.0	11.3	11.9	9.48	7.76	6.49	5.54	4.39	3.60	2.57	1.96	0		
26	0.12	0.28	0.52	0.97	1.81	2.61	3.38	4.14	5.60	7.02	7.72	9.10	10.5	11.8	12.7	10.1	8.21	6.89	5.88	4.66	3.82	2.73	2.08	0		
28	0.13	0.30	0.56	1.05	1.96	2.83	3.67	4.48	6.07	7.61	8.36	9.86	11.3	12.8	14.2	11.2	9.18	7.68	6.57	5.22	4.27	3.05	2.32	0		
30	0.14	0.33	0.61	1.13	2.12	3.05	3.95	4.83	6.54	8.20	9.01	10.6	12.2	13.8	15.3	12.5	10.1	8.51	7.28	5.78	4.73	3.39	2.57	0		
32	0.15	0.35	0.65	1.22	2.27	3.27	4.24	5.18	7.01	8.79	9.66	11.4	13.1	14.7	16.4	13.7	11.2	9.40	8.06	6.37	5.22	3.73	0			
35	0.17	0.38	0.72	1.34	2.50	3.60	4.67	5.70	7.72	9.68	10.6	12.5	14.4	16.2	18.1	15.7	12.8	10.7	9.18	7.28	5.96	4.27	0			
40	0.19	0.44	0.83	1.55	2.89	4.16	5.39	6.59	8.92	11.2	12.3	14.5	16.6	18.8	20.9	19.2	15.7	13.1	11.2	8.88	7.28	5.22	0			
45	0.22	0.50	0.94	1.76	3.28	4.72	6.12	7.48	10.1	12.7	14.0	16.5	18.9	21.3	23.7	22.8	18.7	15.7	13.4	10.6	8.73	0				

Notes: 1. Multiply the value given above by the multi-strand factor in order to obtain the transmission kW of multi-strand chain.

Table B-4

Multi-Strand Factor

Number of Roller Chain Strands	Multi-Strand Factor
2	1.7
3	2.5
4	3.3
5	3.9
6	4.6

Lubricating Methods

A	Manual lubrication or drip lubrication
B	Oil bath lubrication or lubrication by sling disc
C	Lubrication using a pump

NO. 80 MAXIMUM KILOWATT RATINGS

No. of Teeth Small Spkt.	Maximum r/min — Small Sprocket																											
	10	25	50	100	150	200	300	400	500	600	700	800	900	1000	1100	1200	1400	1600	1800	2000	2200	2400	2700	3000	3400			
	Lubrication System																											
	A							B														C						
9	0.32	0.73	1.36	2.53	3.64	4.72	6.80	8.81	10.8	12.7	14.6	15.2	12.7	10.8	9.40	8.21	6.54	5.35	4.48	3.83	3.32	2.91	2.44	2.08	1.73			
10	0.36	0.81	1.52	2.83	4.08	5.29	7.62	9.87	12.1	14.2	16.3	17.8	14.8	12.7	11.0	9.62	7.88	6.27	5.25	4.48	3.89	3.41	2.86	2.44	2.02			
11	0.40	0.90	1.68	3.14	4.52	5.86	8.44	10.9	13.4	15.8	18.1	20.4	17.2	14.6	12.7	11.1	8.80	7.23	6.06	5.17	4.48	3.93	3.30	2.81	1.27			
12	0.43	0.99	1.85	3.45	4.97	6.44	9.27	12.0	14.7	17.3	19.9	22.4	19.5	16.6	14.5	12.7	10.1	8.21	6.90	5.89	5.11	4.48	3.76	3.21	0			
13	0.47	1.08	2.02	3.76	5.42	7.02	10.1	13.1	16.0	18.9	21.7	24.4	22.2	18.8	16.3	14.3	11.3	9.33	7.76	6.65	5.76	5.08	4.24	3.62	0			
14	0.51	1.17	2.18	4.08	5.87	7.60	11.0	14.2	17.3	20.4	23.5	26.5	24.8	21.0	18.2	16.0	12.7	10.4	8.73	7.43	6.44	5.65	4.74	4.04	0			
15	0.55	1.26	2.35	4.39	6.32	8.19	11.8	15.3	18.7	22.0	25.3	28.5	27.5	23.3	20.2	17.8	14.1	11.5	9.62	8.21	7.14	6.27	5.25	4.48	0			
16	0.59	1.35	2.52	4.71	6.78	8.78	12.7	16.4	20.0	23.6	27.1	30.6	30.3	25.7	22.2	19.5	15.5	12.7	10.6	9.10	7.83	6.90	5.79	4.94	0			
17	0.63	1.44	2.69	5.03	7.24	9.38	13.5	17.5	21.4	25.2	29.0	32.7	33.0	28.1	24.4	21.4	16.9	13.9	11.6	9.92	8.58	7.54	6.33	5.41	0			
18	0.67	1.54	2.86	5.35	7.70	9.98	14.4	18.6	22.8	26.8	30.8	34.7	35.9	30.7	26.6	23.3	18.5	15.1	12.7	10.8	9.40	8.21	6.90	5.89	0			
19	0.71	1.63	3.04	5.67	8.16	10.6	15.2	19.7	24.1	28.4	32.7	36.8	39.0	33.2	28.8	25.3	20.1	16.4	13.7	11.7	10.1	8.95	7.46	6.39	0			
20	0.75	1.72	3.21	5.99	8.63	11.2	16.1	20.9	25.5	30.0	34.5	38.9	42.1	35.9	31.1	27.3	21.6	17.8	14.8	12.7	11.0	9.62	8.08	0				
21	0.79	1.81	3.38	6.31	9.10	11.8	17.0	22.0	26.9	31.7	36.4	41.0	45.3	38.6	33.4	29.4	23.3	19.1	16.0	13.7	11.9	10.4	8.73	0				
22	0.84	1.91	3.56	6.64	9.56	12.4	17.8	23.1	28.3	33.3	38.3	43.1	48.0	41.4	35.9	31.5	25.0	20.4	17.2	14.6	12.7	11.1	9.33	0				
23	0.88	2.00	3.73	6.97	10.0	13.0	18.7	24.3	29.7	34.9	40.1	45.3	50.3	44.2	38.3	33.6	28.7	21.9	18.4	15.7	13.6	11.9	10.0	0				
24	0.92	2.09	3.91	7.29	10.5	13.6	19.6	25.4	31.0	36.6	42.0	47.4	52.7	47.5	40.9	35.9	28.5	23.3	19.5	16.6	14.5	12.7	10.6	0				
25	0.96	2.19	4.08	7.62	11.0	14.2	20.5	26.5	32.4	38.2	43.9	49.5	55.1	50.5	43.2	38.1	30.3	24.8	20.7	17.8	15.4	13.5	11.3	0				
26	1.00	2.28	4.26	7.95	11.5	14.8	21.4	27.7	33.9	39.9	45.8	51.7	57.5	53.5	46.1	40.4	32.1	26.3	22.0	18.8	16.3	14.3	12.0	0				
28	1.08	2.47	4.62	8.62	12.4	16.1	23.2	30.0	36.7	43.2	49.6	56.0	62.2	59.8	51.5	45.2	35.9	29.4	24.6	21.0	18.2	16.0						
30	1.17	2.67	4.97	9.28	13.4	17.3	24.9	32.3	39.5	46.6	53.5	60.3	67.1	66.3	57.2	50.1	39.8	32.5	27.3	23.3	20.2	17.8						
32	1.25	2.86	5.33	9.95	14.3	18.6	26.7	34.7	42.4	49.9	57.3	64.7	71.9	72.7	62.9	55.2	43.8	35.9	30.1	25.7	22.2	19.5	0					
35	1.38	3.15	5.87	11.0	15.8	20.5	29.5	38.2	46.7	55.0	63.2	71.2	79.2	83.2	72.0	63.2	50.1	41.0	34.4	29.4	25.4	0	0					
40	1.59	3.64	6.79	12.7	18.2	23.6	34.0	44.1	53.9	63.5	73.0	82.3	91.5	101	87.3	76.8	61.3	50.1	42.0	35.9	14.9	0	0					
45	1.81	4.13	7.71	14.4	20.7	26.8	38.7	50.1	61.2	72.1	82.9	93.4	104	114	106	91.8	73.1	59.8	50.1	40.4	0							

NO. 100 MAXIMUM KILOWATT RATINGS

No. of Teeth Small Spkt.	Maximum r/min - Small Sprocket																										
	10	25	50	100	150	200	300	400	500	600	700	800	900	1000	1100	1200	1300	1400	1600	1800	2000	2200	2400	2600	2700		
	Lubrication System																										
	A						B											C									
9	0.53	1.21	2.26	4.21	6.07	7.86	11.3	14.7	17.9	21.1	22.1	18.1	15.1	13.0	11.2	9.85	8.73	7.83	6.39	5.36	4.57	3.97	3.48	3.09	0		
10	0.59	1.36	2.53	4.72	6.80	8.81	12.7	16.4	20.1	23.7	25.9	21.2	17.9	15.1	13.1	11.6	10.2	9.18	7.46	6.28	5.38	4.65	4.08	3.62	0		
11	0.66	1.50	2.81	5.23	7.54	9.77	14.1	18.2	22.3	26.3	29.9	24.5	20.5	17.5	15.1	13.3	11.8	10.6	8.65	7.24	6.19	5.36	4.71	0.96	0		
12	0.72	1.65	3.08	5.75	8.28	10.7	15.5	20.0	24.5	28.8	33.1	27.8	23.4	19.9	17.3	15.1	13.4	12.0	9.85	8.28	7.05	6.11	5.36	0	0		
13	0.79	1.80	3.36	6.27	9.03	11.7	16.9	21.8	26.7	31.4	36.1	31.4	26.3	22.5	19.5	17.1	15.1	13.6	11.1	9.33	7.91	6.89	6.04	0	0		
14	0.86	1.95	3.64	6.79	9.78	12.7	18.3	23.7	28.9	34.1	39.1	35.1	29.4	25.1	21.8	19.1	16.9	15.1	12.4	10.1	8.88	7.68	6.75	0	0		
15	0.92	2.10	3.92	7.32	10.5	13.7	19.7	25.5	31.1	36.7	42.2	38.9	32.6	27.8	24.2	21.2	18.8	16.8	13.7	11.6	9.85	8.51	7.46	0	0		
16	0.99	2.25	4.20	7.85	11.3	14.6	21.1	27.3	33.4	39.4	45.2	42.9	36.0	30.7	26.6	23.4	20.7	18.5	15.1	12.7	10.8	9.40	8.28	0	0		
17	1.05	2.41	4.49	8.38	12.1	15.6	22.5	29.2	35.7	42.0	48.3	47.0	39.4	33.6	29.1	25.6	22.7	20.3	18.1	14.0	11.9	10.3	0.59	0	0		
18	1.12	2.56	4.77	8.91	12.8	16.6	23.9	31.0	37.9	44.7	51.3	51.2	42.9	36.6	31.7	27.8	24.7	22.1	18.1	15.1	13.0	11.2	0	0	0		
19	1.19	2.71	5.06	9.45	13.6	17.6	25.4	32.9	40.2	47.4	54.4	55.6	46.5	39.7	34.4	30.2	26.8	23.9	19.6	16.4	14.0	12.2	0	0	0		
20	1.26	2.87	5.35	10.0	14.4	18.6	26.8	34.8	42.5	50.1	57.5	60.0	50.2	42.9	37.2	32.6	28.9	25.9	21.2	17.8	15.1	13.1	0	0	0		
21	1.32	3.02	5.64	10.5	15.2	19.6	28.3	36.6	44.8	52.8	60.6	64.6	54.0	46.1	40.0	35.1	31.1	27.8	22.8	19.1	16.3	14.2	0	0	0		
22	1.39	3.18	5.93	11.1	15.9	20.6	29.7	38.5	47.1	55.5	63.8	69.2	58.0	49.5	42.9	37.6	33.4	29.8	24.5	20.5	17.5	15.1	0	0	0		
23	1.46	3.33	6.22	11.6	16.7	21.7	31.2	40.4	49.4	58.2	66.9	74.0	61.9	52.9	46.8	40.2	35.7	31.9	26.1	21.9	18.7	5.77	0	0	0		
24	1.53	3.49	6.51	12.2	17.5	22.7	32.7	42.3	51.7	61.0	70.0	78.9	66.0	56.4	48.9	42.9	38.1	34.0	27.8	23.4	19.9	0	0	0	0		
25	1.60	3.65	6.81	12.7	18.3	23.7	34.1	44.2	54.1	63.7	73.2	82.6	70.2	59.9	51.9	45.6	40.4	36.2	29.6	24.8	21.2	0	0	0	0		
26	1.67	3.81	7.10	13.3	19.1	24.7	35.6	46.2	56.4	66.5	76.4	86.1	74.5	63.6	55.1	48.3	42.9	38.3	31.4	26.3	22.5	0	0	0	0		
28	1.81	4.12	7.69	14.4	20.7	26.8	38.6	50.0	61.1	72.0	82.7	93.3	83.6	71.0	61.6	54.0	47.9	42.9	35.1	29.4	25.1	0	0	0	0		
30	1.95	4.44	8.29	15.5	22.3	28.9	41.6	53.9	65.8	77.6	89.1	101	92.5	79.1	68.3	59.9	53.1	47.5	38.9	32.6	7.46	0	0	0	0		
32	2.09	4.76	8.89	16.6	23.9	31.0	44.6	57.8	70.6	83.2	95.6	108	101	86.5	75.4	66.0	58.6	52.4	42.9	33.7	0	0	0	0	0		
35	2.30	5.25	9.79	18.3	26.3	34.1	49.1	63.6	77.8	91.6	105	119	116	99.2	85.8	75.4	67.0	59.9	49.1	41.1	0	0	0	0	0		
40	2.66	6.06	11.3	21.1	30.4	39.4	56.7	73.5	89.8	106	122	137	142	122	105	92.5	82.1	73.2	59.9	0	0	0	0	0	0		
45	3.02	6.68	12.8	24.0	34.5	44.7	64.4	83.5	102	120	138	156	170	145	125	110	97.7	87.3	33.8	0	0	0	0	0	0		